

Solve the absolute value problems for x .

1. $|2x - 9| = 17$

Solution:

$$\begin{aligned} |2x - 9| = 17 &\implies 2x - 9 = \pm 17 \\ &\implies 2x = 26 \quad \text{or} \quad 2x = -8 \\ &\implies x = 13 \quad \text{or} \quad x = -4 \\ &\implies \boxed{x = -4, 13} \end{aligned}$$

2. $|5x - 1| = 3x$

Solution:

$$\begin{aligned} |5x - 1| = 3x &\implies 5x - 1 = \pm 3x \\ &\implies 5x - 1 = 3x \quad \text{or} \quad 5x - 1 = -3x \\ &\implies \boxed{x = \frac{1}{2}, \frac{1}{8}} \end{aligned}$$

3. $|3x| = |6 - x|$

Solution:

$$\begin{aligned} |3x| = |6 - x| &\implies \left| \frac{3x}{6 - x} \right| = 1 \implies \frac{3x}{6 - x} = \pm 1 \\ &\implies 3x = 6 - x \quad \text{or} \quad 3x = -(6 - x) \\ &\implies \boxed{x = \frac{3}{2}, -3} \end{aligned}$$

4. $|2x - 5| + |x + 1| = 8$

Solution: Critical points: $2x - 5 = 0 \Rightarrow x = \frac{5}{2}, \quad x + 1 = 0 \Rightarrow x = -1$

Break into intervals: $x < -1, -1 \leq x < \frac{5}{2}, x \geq \frac{5}{2}$

Case 1: $x < -1$

$$|2x - 5| + |x + 1| = -(2x - 5) - (x + 1) = -3x + 4 = 8 \Rightarrow x = -\frac{4}{3}$$

Case 2: $-1 \leq x < \frac{5}{2}$

$$|2x - 5| + |x + 1| = -(2x - 5) + (x + 1) = -x + 6 = 8 \Rightarrow x = -2 \quad \text{Not in range}$$

Case 3: $x \geq \frac{5}{2}$

$$|2x - 5| + |x + 1| = (2x - 5) + (x + 1) = 3x - 4 = 8 \Rightarrow x = 4$$

Final Answer:

$$\boxed{x = -\frac{4}{3} \text{ or } x = 4}$$

5. $x^2 = x\sqrt{u^2}$

Solution:

$$x^2 = x\sqrt{u^2} \implies x^2 - x\sqrt{u^2} = 0$$

$$\implies x(x - |u|) = 0$$

$$\implies \boxed{x = 0, |u|}$$